

Section 7.6 Atthaseth Kanokwuthithammong 125010006 T03

$$1. a) \tan^{-1} 1 = \frac{\pi}{4}$$

$$b) \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$$

$$c) \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$3. a) \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

$$b) \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

$$c) \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$$

$$5. a) \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$$b) \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{3\pi}{4}$$

$$c) \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

$$7. a) \sec^{-1}(-\sqrt{2}) = \frac{3\pi}{4}$$

$$b) \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$c) \sec^{-1}(-2) = \frac{2\pi}{3}$$

$$13. \lim_{x \rightarrow 1} \sin^{-1} X = \sin^{-1} 1 = \frac{\pi}{2}$$

$$16. \lim_{x \rightarrow -\infty} \tan^{-1} X = -\frac{\pi}{2}$$

$$18. \lim_{x \rightarrow -\infty} \sec^{-1} X = \lim_{x \rightarrow 0} \cos^{-1} X = \cos^{-1} 0 = \frac{\pi}{2}$$

$$29. y = \cos^{-1}\left(\frac{1}{x}\right) \rightarrow \frac{dy}{dx} = -\frac{1}{\sqrt{1-\left(\frac{1}{x}\right)^2}} \cdot \left(-\frac{1}{x^2}\right) = \frac{1}{\sqrt{x^4-x^2}}$$

$$25. y = \sec^{-1}(2s+1) \rightarrow \frac{dy}{ds} = \frac{1}{|2s+1|\sqrt{(2s+1)^2-1}} \cdot 2 = \frac{1}{|2s+1|\sqrt{s^2+s}}$$

$$33. y = \ln(\tan^{-1} X) \rightarrow \frac{dy}{dx} = \frac{1}{\tan^{-1} X} \cdot \left(\frac{1}{1+X^2}\right)$$

$$44. u = 2x \rightarrow du = 2dx \therefore \int \frac{dx}{\sqrt{1-4x^2}} = \int \frac{1}{2} \left(\frac{1}{\sqrt{1-u^2}} \right) du = \frac{1}{2} (\sin^{-1} u) + C = \frac{1}{2} \sin^{-1}(2x) + C$$

$$47. u = \frac{5}{\sqrt{2}}x \rightarrow du = \frac{5}{\sqrt{2}}dx \therefore \int \frac{dx}{x\sqrt{25x^2+2}} = \int \frac{\frac{1}{5} du}{\frac{1}{5} u \sqrt{u^2-1}} = \int \frac{1}{u \sqrt{u^2-1}} du = \frac{1}{\sqrt{2}} \sec^{-1} u + C = \frac{1}{\sqrt{2}} \sec^{-1}\left(\frac{5\sqrt{2}}{2}x\right) + C$$

$$52. u = \frac{9}{2}t \rightarrow du = \frac{9}{2}dt \therefore \int \frac{dt}{4+9t^2} = \int \frac{1}{4+u^2} \cdot \frac{2}{9} du = \frac{2}{9} \int \frac{1}{4+u^2} du = \frac{2}{9} \left[\tan^{-1} \frac{u}{2} \right]_{-\sqrt{3}}^{\sqrt{3}} = \frac{2}{9} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right) = \frac{2\pi}{9}$$

$$63. u = e^x \rightarrow du = e^x dx \therefore \int_0^{\ln 5} \frac{e^x dx}{1+e^{2x}} = \int_1^5 \frac{du}{1+u^2} = \left[\tan^{-1} u \right]_1^5 = \frac{\pi}{3} - 0 = \frac{\pi}{3}$$

$$65. u = y^2 \rightarrow du = 2y dy \therefore \int \frac{y dy}{\sqrt{1-y^4}} = \int \frac{\frac{1}{2} du}{\sqrt{1-u^2}} = \frac{1}{2} (\sin^{-1} u) + C = \frac{1}{2} \sin^{-1}(y^2) + C$$

$$69. u = t+1 \rightarrow du = dt, 3-2t-t^2 = 4-(t+1)^2 = 4-u^2 \therefore \int \frac{6 dt}{\sqrt{3-2t-t^2}} = \int \frac{6 du}{\sqrt{4-u^2}} = \left[6 \sin^{-1} \frac{u}{2} \right]_0^1 = 6 \cdot \frac{\pi}{6} - 6 \cdot 0 = \pi$$

$$74. u = t-3 \rightarrow du = dt, t^2-6t+10 = (t-3)^2+1 = u^2+1 \therefore \int \frac{2 dx}{x^2-6x+10} = \int \frac{2 du}{u^2+1} = \left[2 \tan^{-1} u \right]_{-1}^1 = 2\left(\frac{\pi}{4}\right) - 2\left(-\frac{\pi}{4}\right) = \pi$$

$$87. u = \sec^{-1} X \rightarrow du = \frac{1}{|X|\sqrt{X^2-1}} dX \therefore \int_{\sqrt{2}}^2 \frac{\sec^{-1}(X)}{X\sqrt{X^2-1}} dX = \int_{\pi/4}^{\pi/3} \sec^{-1} u \cdot du = \left[\tan u \right]_{\pi/4}^{\pi/3} = \sqrt{3} - 1$$

$$91. \lim_{x \rightarrow 0} \frac{\sin^{-1} 5x}{x} = \lim_{x \rightarrow 0} \frac{5/\sqrt{1-25x^2}}{1} = \frac{5}{\sqrt{1-0}} = 5$$

$$92. \lim_{x \rightarrow 1^+} \frac{\sqrt{x^2-1}}{\sec^{-1} X} = \lim_{x \rightarrow 1^+} \frac{\frac{1}{2\sqrt{x^2-1}} \cdot 2x}{\frac{1}{|x|\sqrt{x^2-1}}} = \lim_{x \rightarrow 1^+} \frac{x}{|x|} = \frac{1}{1} = 1$$

$$96. \lim_{x \rightarrow \infty} \frac{e^x \tan^{-1} e^x}{e^{2x} + x} = \lim_{x \rightarrow \infty} \frac{\frac{e^{2x}}{1+e^{2x}} + e^x \tan^{-1} e^x}{2e^{2x} + 1} = \lim_{x \rightarrow \infty} \frac{2e^x \tan^{-1} e^x + \frac{e^x}{e^{2x}+1} + \frac{2e^{2x}}{(e^{2x}+1)^2}}{4e^{2x}} =$$

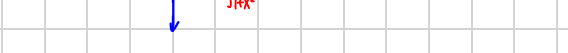
$$= \lim_{x \rightarrow \infty} \left(\frac{\tan^{-1} e^x}{2e^{2x}} + \frac{1}{4e^{2x}(e^{2x}+1)} + \frac{1}{2(e^{2x}+1)^2} \right) = 0 + 0 + 0 =$$

$$97. \lim_{x \rightarrow 0^+} \frac{(\tan^{-1} \sqrt{x})^2}{x\sqrt{x+1}} = \lim_{x \rightarrow 0^+} \frac{2(\tan^{-1} \sqrt{x}) \left(\frac{1}{1+x} \right) \left(\frac{1}{2\sqrt{x}} \right)}{\sqrt{x+1} + \frac{x}{2\sqrt{x+1}}} = \lim_{x \rightarrow 0^+} \left(\frac{1}{\sqrt{x+1}} \right) \cdot \lim_{x \rightarrow 0^+} \frac{\tan^{-1} \sqrt{x}}{\sqrt{x}} = \frac{1}{1} \cdot \lim_{x \rightarrow 0^+} \frac{\frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}}}{\frac{1}{2\sqrt{x}}} = \frac{1}{1} \cdot \frac{1}{1+0} = 1$$

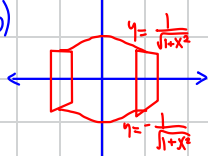
$$123. a) \text{Volume} = -\int_1^{\sqrt{2}} \pi y^2 dx$$

$$= -\int_1^{\sqrt{2}} \pi \left(\frac{1}{1+x^2} \right) dx$$

$$= \pi \left[\tan^{-1} X \right]_{-1}^1 = \pi \left(\frac{\pi}{4} - \left(-\frac{\pi}{4}\right) \right) = \frac{\pi^2}{2}$$



123. b)



$$\begin{aligned} \text{Volume} &= \int_{-1}^1 (2y)^2 dx \\ &= \int_{-1}^1 4 \cdot \frac{1}{1+x^2} dx \\ &= 4 [\arctan x]_{-1}^1 = 4 \left(\frac{\pi}{4} - \left(-\frac{\pi}{4}\right) \right) = 2\pi \end{aligned}$$

127. a) $y = \tan^{-1}(\tan x)$

Domain $(y) = (-\infty, \infty)$

except those in form $\frac{\pi}{2} + k\pi$ ($k \in \mathbb{Z}$)

Range $(y) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

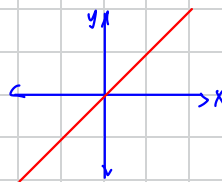
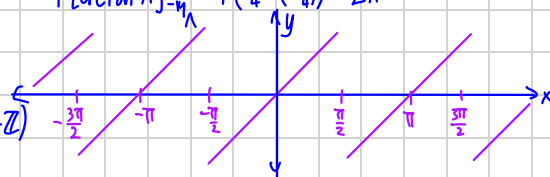
→ The function is periodic ($\tan^{-1}x$ has range of $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$)

b) $y = \tan(\tan^{-1}x) = x$

Domain $(y) = (-\infty, \infty)$

Range $(y) = (-\infty, \infty)$

→ Make sense: the inverse function turns the value back to original value



Section 8.2

2. $u = \theta \rightarrow du = d\theta$, $v = \frac{\sin \pi \theta}{\pi} \rightarrow dv = \cos \pi \theta d\theta$

$$\therefore \int \theta \cos \pi \theta d\theta = \frac{\theta \sin \pi \theta}{\pi} - \int \frac{\sin \pi \theta}{\pi} d\theta = \frac{\theta \sin \pi \theta}{\pi} + \frac{\cos \pi \theta}{\pi^2} + C$$

6. $u = \ln x \rightarrow du = \frac{1}{x} dx$, $v = \frac{x^4}{4} \rightarrow dv = x^3 dx$

$$\therefore \int x^3 \ln x dx = \left[\frac{x^4}{4} \ln x \right] - \int \frac{x^4}{4} dx = \frac{e^4}{4} \cdot 1 - \frac{1}{4} \cdot 0 - \left[\frac{x^4}{16} \right]_1 = \frac{e^4}{4} - \frac{e^4}{16} + \frac{1}{16} = \frac{3e^4 + 1}{16}$$

12. $u = \sin^{-1} y \rightarrow du = \frac{1}{\sqrt{1-y^2}} dy$, $v = y \rightarrow dv = dy$

$$\int \sin^{-1} y dy = y \sin^{-1} y - \int \frac{y}{\sqrt{1-y^2}} dy = y \sin^{-1} y + \sqrt{1-y^2} + C$$

17. $u = x^2 - 5x \rightarrow du = 2x - 5$, $v = e^x \rightarrow dv = e^x dx$

$$\therefore \int (x^2 - 5x) e^x dx = (x^2 - 5x) e^x - \int (2x - 5) e^x dx = (x^2 - 5x) e^x - \int 2x e^x dx + \int 5e^x dx = (x^2 - 5x) e^x - 2 \int x e^x dx + 5e^x + C$$

for $\int x e^x$, $u = x \rightarrow du = dx$, $v = e^x \rightarrow dv = e^x dx$ $\therefore \int x e^x = x e^x - \int e^x dx = x e^x - e^x + C$

$$\therefore \int (x^2 - 5x) e^x dx = (x^2 - 5x) e^x - 2(x e^x - e^x) + 5e^x + C = e^x (x^2 - 7x + 7) + C$$

23. $u = e^{2x} \rightarrow du = 2e^{2x} dx$, $v = \frac{\sin 3x}{3} \rightarrow dv = \cos 3x dx$

$$\therefore \int e^{2x} \cos 3x dx = \frac{e^{2x} \sin 3x}{3} - \int \frac{2}{3} e^{2x} \sin 3x dx = \frac{e^{2x} \sin 3x}{3} - \frac{2}{3} \int e^{2x} \sin 3x dx \quad \text{--- (1)}$$

$$u = e^{2x} \rightarrow du = 2e^{2x} dx$$
, $v = -\frac{\cos 3x}{3} \rightarrow dv = \sin 3x dx$

$$\therefore \int e^{2x} \sin 3x dx = \frac{-e^{2x} \cos 3x}{3} - \int \frac{2}{3} e^{2x} \cos 3x dx = \frac{-e^{2x} \cos 3x}{3} + \frac{2}{3} \int e^{2x} \cos 3x dx \quad \text{--- (2)}$$

Plug (2) in (1) $\int e^{2x} \cos 3x dx = \frac{e^{2x} \sin 3x}{3} - \frac{2}{3} \left(\frac{-e^{2x} \cos 3x}{3} + \frac{2}{3} \int e^{2x} \cos 3x dx \right)$

$$= \frac{e^{2x} \sin 3x}{3} + \frac{2e^{2x} \cos 3x}{9} - \frac{4}{9} \int e^{2x} \cos 3x dx + C$$

$$\therefore \frac{13}{9} \int e^{2x} \cos 3x dx = \frac{e^{2x}}{9} (3 \sin 3x + 2 \cos 3x) + C$$

$$\rightarrow \int e^{2x} \cos 3x dx = \frac{e^{2x}}{13} (3 \sin 3x + 2 \cos 3x) + C$$

28. $u = \ln(x+x^2) \rightarrow du = \frac{2x+1}{x+x^2} dx$, $v = x \rightarrow dv = dx$
 $\int \ln(x+x^2) dx = x \ln(x+x^2) - \int \frac{2x+1}{x+x^2} dx$
 $\therefore \int \frac{2x+1}{x+x^2} dx = \int \frac{x(2x+1)}{x(x+1)} dx = \int (2 - \frac{1}{x+1}) dx = 2x - \ln(x+1) + C$
 $\therefore \int \ln(x+x^2) dx = x \ln(x+x^2) - 2x + \ln(x+1) + C$ #

29. $u = \sin(\ln x) \rightarrow du = \cos(\ln x) \cdot \frac{1}{x}$, $v = x \rightarrow dv = dx$
 $\therefore \int \sin(\ln x) dx = x \sin(\ln x) - \int \cos(\ln x) dx$ — ①
 $u = \cos(\ln x) \rightarrow du = -\sin(\ln x) \cdot \frac{1}{x}$, $v = x \rightarrow dv = dx$
 $\therefore \int \cos(\ln x) dx = x \cos(\ln x) - \int -\sin(\ln x) dx = x \cos(\ln x) + \int \sin(\ln x) dx$ — ②

Plug ② in ①: $\int \sin(\ln x) dx = x \sin(\ln x) - x \cos(\ln x) - \int \sin(\ln x) dx$
 $\therefore 2 \int \sin(\ln x) dx = x(\sin(\ln x) - \cos(\ln x)) + C \rightarrow \int \sin(\ln x) dx = \frac{x}{2}(\sin(\ln x) - \cos(\ln x)) + C$ #

31. $u = x^2 \rightarrow du = 2x dx$ $\therefore \int x \sec x^2 dx = \int \frac{1}{2} \sec u du = \frac{1}{2} \ln |\sec u + \tan u| + C = \frac{1}{2} \ln |\sec x^2 + \tan x^2| + C$

32. $u = \sqrt{x} \rightarrow du = \frac{1}{2\sqrt{x}} dx$ $\therefore \int \frac{\cos \sqrt{x}}{\sqrt{x}} dx = \int 2 \cos u du = 2 \sin u + C = 2 \sin \sqrt{x} + C$

37. $u = x^4 \rightarrow du = 4x^3 dx$ $\therefore \int x^3 e^{x^4} dx = \int \frac{1}{4} e^u du = \frac{1}{4} e^u + C = \frac{1}{4} e^{x^4} + C$

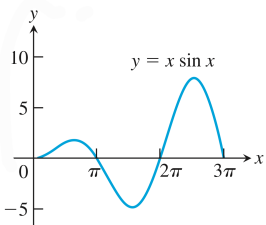
43. $u = \ln x \rightarrow du = \frac{1}{x} dx$, $v = \frac{2}{3} x^{3/2} \rightarrow dv = \sqrt{x} dx$
 $\therefore \int \sqrt{x} \ln x dx = \frac{2}{3} x^{3/2} \ln x - \int \frac{2}{3} x^{3/2} \cdot \frac{1}{x} dx = \frac{2}{3} x^{3/2} \ln x - \frac{2}{3} \int \sqrt{x} dx = \frac{2}{3} x^{3/2} \ln x - \frac{2}{3} (\frac{2}{3} x^{3/2}) + C = \frac{2}{3} x^{3/2} \ln x - \frac{4}{9} x^{3/2} + C$

45. $u = 2\sqrt{x} \rightarrow du = \frac{1}{\sqrt{x}} dx$, $v = \sin \sqrt{x} \rightarrow dv = \frac{\cos \sqrt{x}}{2\sqrt{x}} dx$
 $\therefore \int \cos \sqrt{x} dx = 2\sqrt{x} \sin \sqrt{x} - \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = 2\sqrt{x} \sin \sqrt{x} + 2 \cos \sqrt{x} + C$ (From Q12)

50. $u = x^2 \rightarrow du = 2x dx$ $\therefore \int_0^{1/\sqrt{2}} 2x \sin^{-1}(x^2) dx = \int_0^{1/2} \sin^{-1}(u) du = [X \sin^{-1} X + \sqrt{1-X^2}]_0^{1/2} = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$

51. $u = \tan^{-1} x \rightarrow du = \frac{1}{1+x^2} dx$, $v = \frac{x^2}{2} \rightarrow dv = x dx$
 $\therefore \int x \tan^{-1} x dx = \frac{x^2 \tan^{-1} x}{2} - \int \frac{x^2}{2(1+x^2)} dx = \frac{x^2 \tan^{-1} x}{2} - \frac{1}{2} \int (1 - \frac{1}{1+x^2}) dx = \frac{x^2 \tan^{-1} x}{2} - \frac{1}{2} (x - \tan^{-1} x) + C$

53. $a) u = x \rightarrow du = dx$, $v = -\cos x \rightarrow dv = \sin x dx$
 $Area = \int_0^\pi x \sin x dx = [-x \cos x + \int \cos x dx]_0^\pi = [-x \cos x + \sin x]_0^\pi = \pi - 0 = \pi$
 $b) Area = -\int_\pi^{2\pi} x \sin x dx = [-x \cos x + \sin x]_\pi^{2\pi} = -(-2\pi - \pi) = 3\pi$
 $c) Area = \int_{2\pi}^{3\pi} x \sin x dx = [-x \cos x + \sin x]_{2\pi}^{3\pi} = 3\pi - (-2\pi) = 5\pi$
 $d) Area \text{ between the curve and } x\text{-axis for } n\pi \leq x \leq (n+1)\pi; n \in \mathbb{N}$
 $= \left| \int_{n\pi}^{(n+1)\pi} x \sin x dx \right| = \left| [-x \cos x + \sin x]_{n\pi}^{(n+1)\pi} \right| = \left| -(n+1)\pi \cos[(n+1)\pi] + n\pi \cos(n\pi) \right|$
 $= (n+1)\pi + n\pi = (2n+1)\pi$



61. Average value of $y = \frac{1}{2\pi} \int_0^{2\pi} 2e^{-t} \cos t dt = \frac{1}{\pi} \int_0^{2\pi} e^{-t} \cos t dt$ — ①

$u = e^{-t} \rightarrow du = -e^{-t} dt$, $v = \sin t \rightarrow dv = \cos t dt$

$\therefore \int e^{-t} \cos t dt = e^{-t} \sin t + \int e^{-t} \sin t dt$ — ①

$u = e^{-t} \rightarrow du = -e^{-t} dt$, $v = -\cos t \rightarrow dv = \sin t dt$

$\therefore \int e^{-t} \sin t dt = -e^{-t} \cos t - \int e^{-t} \cos t dt$ — ②

Plug ② in ①: $\int e^{-t} \cos t dt = e^{-t} \sin t - e^{-t} \cos t - \int e^{-t} \cos t dt$

$\therefore 2 \int e^{-t} \cos t dt = e^{-t} \sin t - e^{-t} \cos t \therefore \int e^{-t} \cos t dt = \frac{e^{-t}}{2} (\sin t - \cos t)$

Plug into ①: Average value of $y = \frac{1}{\pi} \int_0^{2\pi} e^{-t} \cos t dt = \frac{1}{\pi} \left[\frac{e^{-t}}{2} (\sin t - \cos t) \right]_0^{2\pi} = \frac{1}{2\pi} (1 - e^{-2\pi})$

63. $u = x^n \rightarrow du = nx^{n-1} dx$, $v = \sin x \rightarrow dv = \cos x dx$

$\therefore \int x^n \cos x dx = x^n \sin x - \int \sin x \cdot nx^{n-1} dx = x^n \sin x - n \int x^{n-1} \sin x dx$

69. Let $F'(x) = f(x)$, $u = x-a \rightarrow du = dx$, $v = \int_a^x f(t) dt = -\int_x^a f(t) dt \rightarrow dv = f(x) dx$

$\therefore \int_a^b (x-a)f(x) dx = \left[(x-a) \int_a^x f(t) dt \right]_a^b - \int_a^b \left(\int_a^x f(t) dt \right) dx$
 $= (0-0) + \int_a^b \left(\int_x^a f(t) dt \right) dx = \int_a^b \left(\int_x^a f(t) dt \right) dx$

Section 8.3

3. $\int \cos^3 x \sin x dx = \int -(\cos^3 x)(d(\cos x)) = -\frac{\cos^4 x}{4} + C$

14. $\int_0^{\pi/2} \sin^2 x dx = \int_0^{\pi/2} \frac{1-\cos 2x}{2} dx = \left[\frac{x}{2} - \frac{\sin 2x}{4} \right]_0^{\pi/2} = \frac{\pi}{4} - 0 = \frac{\pi}{4}$

15. $\int_0^{\pi/2} \sin^3 y dy = \int_0^{\pi/2} (1-\cos^2 y) \sin y dy = \int_0^{\pi/2} (1-3\cos^4 y + 3\cos^2 y - \cos^6 y)(-d(\cos y))$
 $= \int_0^{\pi/2} (1-3u^4+3u^2-u^6) du = \left[u - \frac{3u^5}{5} + \frac{3u^3}{3} - \frac{u^7}{7} \right]_0^{\pi/2}$
 $= \frac{16}{35}$

19. $\int 16 \sin^2 x \cos^3 x dx = \int 16 \left(\frac{1-\cos 2x}{2} \right) \left(\frac{1+\cos 2x}{2} \right) dx = \int 4(1-\cos^2 2x) dx = \int 4 - 4 \left(\frac{1+\cos 4x}{2} \right) dx = \int 2 - 2\cos 4x dx = 2x - \frac{\sin 4x}{2} + C$

20. $\int_0^{\pi} 8 \sin^4 y \cos^2 y dy = \int_0^{\pi} 8 \left(\frac{1-\cos 2y}{2} \right)^2 \left(\frac{1+\cos 2y}{2} \right) dy = \int_0^{\pi} (1-\cos 2y)(1-\cos^2 2y) dy = \int_0^{\pi} (1-\cos 2y - \cos^2 2y + \cos^3 2y) dy$
 $= \left[y - \frac{\sin 2y}{2} - \left(\frac{1}{2} + \frac{\sin 4y}{8} \right) + \left(\frac{\sin 2y}{2} - \frac{\sin^3 2y}{6} \right) \right]_0^{\pi} = \pi - 0 - \frac{\pi}{2} - 0 + 0 - 0 = \frac{\pi}{2}$

24. $\int_0^{\pi} \sqrt{1-\cos 2x} dx = \int_0^{\pi} \sqrt{2-2\cos x} dx = \int_0^{\pi} \sqrt{2} \cdot \sqrt{1-\cos x} dx = \int_0^{\pi} \sqrt{2} \cdot \sin \frac{x}{2} dx = \sqrt{2} [-\cos \frac{x}{2}]_0^{\pi} = 2\sqrt{2}$

27. $\int_{\pi/4}^{\pi/2} \frac{\sin^2 x}{1-\cos x} dx = \int_{\pi/4}^{\pi/2} \frac{1-\cos^2 x}{1-\cos x} \cdot \sin x dx = \int_{\pi/4}^{\pi/2} \frac{1+\cos x}{1-\cos x} \cdot \sin x dx = \int_0^{1/2} \frac{1+u}{1-u} \cdot du = \left[\frac{2}{3}(1-u)^{3/2} \right]_0^{1/2} = \frac{2}{3} \left(\frac{3}{2} \right)^{3/2} - \frac{2}{3}(1) = \sqrt{\frac{3}{2}} - \frac{2}{3} = \frac{3\sqrt{6}-4}{6}$

30. $\int_{\pi/2}^{\pi} \sqrt{1-\sin 2x} dx = \int_{\pi/2}^{\pi} \sqrt{\sin^2 x - 2\sin x \cos x + \cos^2 x} dx = \int_{\pi/2}^{\pi} |\sin x - \cos x| dx = [-\cos x - \sin x]_{\pi/2}^{\pi} = 0 + 1 = 1$

33. $u = \tan x \rightarrow du = \sec^2 x dx$ $\therefore \int \sec^2 x \tan x dx = \int u du = \frac{u^2}{2} + C = \frac{\tan^2 x}{2} + C$

34. $\int \sec x \tan^2 x dx = \int \sec x (\sec^2 x - 1) dx = \int \sec^3 x dx - \int \sec x dx$ — (2)

For $\int \sec^3 x dx$, we use $u = \sec x \rightarrow du = \sec x \tan x dx$, $v = \tan x \rightarrow dv = \sec^2 x dx$

$\therefore \int \sec^3 x dx = \sec x \tan x - \int \sec x \tan^2 x dx$

Plug in (2): $\int \sec x \tan^2 x dx = \sec x \tan x - \int \sec x \tan^2 x dx - \int \sec x dx$

$2 \int \sec x \tan^2 x dx = \sec x \tan x - \ln |\sec x + \tan x| + C \quad \therefore \int \sec x \tan^2 x dx = \frac{1}{2} (\sec x \tan x - \ln |\sec x + \tan x|) + C$

35. $u = \sec x \rightarrow du = \sec x \tan x dx \quad \therefore \int \sec^3 x \tan x dx = \int u^2 du = \frac{u^3}{3} + C = \frac{\sec^3 x}{3} + C$

53. $\int_{-\pi}^{\pi} \sin^2 3x dx = \int_{-\pi}^{\pi} \frac{1-\cos 6x}{2} dx = \left[\frac{x}{2} - \frac{\sin 6x}{12} \right]_{-\pi}^{\pi} = \pi$

54. $\int_{\pi/2}^{\pi} \sin x \cos x dx = \int_{\pi/2}^{\pi} \frac{\sin 2x}{2} dx = \left[-\frac{\cos 2x}{4} \right]_{\pi/2}^{\pi} = \frac{1}{2}$

56. $\int_{-\pi/2}^{\pi/2} \cos x \cos 7x dx = \int_{-\pi/2}^{\pi/2} \frac{\cos 8x + \cos 6x}{2} dx = \left[\frac{\sin 8x}{16} + \frac{\sin 6x}{12} \right]_{-\pi/2}^{\pi/2} = 0$

64. $\int \frac{\sin^2 x}{\cos^3 x} dx = \int \frac{\cos^2 x - 1}{\cos^3 x} (-\sin x) dx = \int \left(\frac{1}{\cos x} - \frac{1}{\cos^3 x} \right) d(\cos x) = -\frac{1}{\cos x} + \frac{1}{3\cos^3 x} + C$

67. $u = x \rightarrow du = dx$, $dv = \sin^2 x dx = \frac{1-\cos 2x}{2} \rightarrow v = \frac{x}{2} - \frac{\sin 2x}{4}$
 $\therefore \int x \sin^2 x dx = x \left(\frac{x}{2} - \frac{\sin 2x}{4} \right) - \int \left(\frac{x}{2} - \frac{\sin 2x}{4} \right) dx = \frac{x^2}{2} - \frac{x \sin 2x}{4} - \left[\frac{x^2}{4} + \frac{\cos 2x}{8} \right] + C = \frac{x^2}{4} - \frac{x \sin 2x}{4} + \frac{\cos 2x}{8} + C$